CCC-C-458a
February 20, 1965
SUPERSEDING
Fed. Spec. CCC-C-458
May 17, 1961 and
Fed. Spec. CCC-F-466
August 19, 1936
(See section 6)

FEDERAL SPECIFICATION

CLOTH, FLANNEL, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton flannels (see 6.1).

1.2 Classification.

1.2.1 *Types and classes.* The finished cloth shall be of the following types and classes, as specified (see 6.2):

Type I—Plain weave (soft filled sheeting), unbleached, napped one side.

Class 1—4.7 ounces.

Type II—Plain weave (soft filled sheeting), unbleached, napped both sides.

Class 1—4.6 ounces.

Type III—Plain weave (outing flannel), bleached or dyed, napped both sides.

Class 1—3.5 ounces.

Class 2—4.0 ounces.

Class 3—4.6 ounces.

Type IV—Twill weave (canton flannel), unbleached, napped on the filling side.

Class 1—5.0 ounces.

Class 2—6.0 ounces.

Class 3—7.0 ounces.

Class 4—8.0 ounces.

Class 5—10.0 ounces.

Class 6—12.0 ounces.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished type III cloth shall match the standard sample for shade; and the unbleached, bleached, or dyed and finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn, and spun into single yarn (see 4.4) for both the warp and the filling.

3.3 Color. The types I, II, and IV cloth shall be unbleached. The type III cloth shall be bleached white or dyed as specified (see 6.2). The use of sulfur dyes or dyes containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4. For civil agencies, when no standard sample is available, there shall be a minimum amount of staining of the test paper when tested as specified in 4.4.

3.3.1 *Matching.* The finished type III cloth shall match the standard sample under natural (north sky) daylight having a color temperature of 7500° Kelvin (K.) and shall be a good approximation to the standard sample under incandescent lamplight at 2800° K.

3.3.2 *Colorfastness.* The dyed and finished cloth shall show colorfastness to laundering, crocking, and perspiration equal to or better than the standard sample. When no standard sample is established, the dyed and finished cloth shall show good colorfastness

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

to laundering and perspiration and shall show a Munsell Value for crocking not lower than 8.5. Tests shall be as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the physical requirements shown in tables I, II, and III when tested as specified in 4.4.

TABLE I. *Gray soft-filled sheeting*

Type	Class	Weight oz./sq. yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
I	1	4.7	42	40	26	21 S/N[1]
II	1	4.6	42	40	26	19 D/N[2]

[1] Single napped.

[2] Double napped.

TABLE II. *Outing flannel*

Type	Class	Weight oz/sq. yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
III	1	3.5	42	40	20	17
III	2	4.0	42	40	23	19
III	3	4.6	50	40	28	23

TABLE III. *Canton flannel*

Type	Class	Weight oz/sq. yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
IV	1	5.0	62	38	45	20
IV	2	6.0	62	38	60	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

TABLE III. *Canton flannel* (continued)

Type	Class	Weight oz/sq. yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
IV	3	7.0	62	38	60	30
IV	4	8.0	62	38	65	40
IV	5	10.0	60	34	100	50
IV	6	12.0	60	34	100	70

3.4.1 *Weave.* The weave for types I, II, and III shall be plain. The weave for type IV shall be a 3 or 4 harness twill.

3.4.2 *Width.* The width shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish. The nap of the finished cloth shall be equal to or better than the standard sample (see 3.1 and 6.3).

3.5.1 *Types.*

3.5.1.1 Type I cloth shall be napped one side.

3.5.1.2 Type II cloth shall be napped both sides.

3.5.1.3. Type III cloth shall be napped both sides.

3.5.1.4 Type IV cloth shall be napped on the filling side.

3.5.2 *Singeing.* When specified (see 6.2), type IV, class 3 cloth shall be singed on the unnapped side.

3.6 Shrinkage. When specified (see 6.2), type I and type III, class 3 cloth shall be preshrunk and shall not shrink more than 2.0 percent in either the warp or the filling direction when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in PPP–P–51.

3.8 Fiber content. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship. The cloth shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection require-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

ments as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL–STD–105 and 4.2.1 through 4.2.4.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the napped face side and all defects as defined in section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1. No one of the 30 contiguous linear yards, folded yards, or increments of 1 yard on the measuring device of the inspection machine, selected for examination from any one piece within the sample, shall be penalized more than four points. The sample unit for this examination shall be one linear yard. The sample size shall be 750 yards, regardless of lot size. The sample yardage shall be examined from 25 pieces taken from 25 containers. An approximately equal number of yards shall be examined in each piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceed the following values:

32.0 points—Types I, II, and IV—unbleached

28.0 points—Type III—bleached

25.0 points—Type III—dyed

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	—	One point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	—	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	—	three points
For defects exceeding 9 inches in any dimension	—	four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Overall uncleanness.

Mottled, streaky, or cloudy (type III only).

Objectionable odor.

Cloth not singed as specified (type IV, class 3 only).

Degree of napping not equal to, or greater than, standard sample.

Width less than specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for shade (type III only).* During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample which is off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for identification of preshrinkage process and compliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP–P–51 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC–T–191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be as follows:

a. Three yards, full width of the finished cloth when preshrunk cloth is specified.

b. One and one half yards, full width of the finished cloth when dyed and unshrunk cloth is specified.

c. One yard, full width of the finished cloth when undyed and unshrunk cloth is specified.

The sample size shall be as shown in the list below. The lot shall be unacceptable if one or more units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE IV. *Test methods*

Characteristic	Requirement reference	Test method
Yarn	3.2	Visual [1]
Weave	3.4.1	Visual [1]
Weight	3.4	5041
Yarns per inch	3.4	5050

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

TABLE IV. *Test methods* (continued)

Characteristic	Requirement reference	Test method
Breaking strength	3.4	5100
Sulfur content	3.3	4.4.1 [1]
Colorfastness to:		
Laundering	3.3.2	5610
Crocking	3.3.2	5651
Perspiration	3.3.2	5680
Shrinkage	3.6	5556

[1] One determination shall be made on each sample unit and the result reported as "pass" or "fail".

4.4.1 *Test for presence of sulfur content.*

4.4.1.1 *Solution (acid stannous chloride).* Dissolve 100 g. of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.1.2 *Procedure.* One to 1.5 g. of the finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous choride solution. Place a filter paper which has been moistened with one drop of 5.0 percent lead acetate solution (made by dissolving 5 g. of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.1.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory. For civil agencies, when no standard sample is available, there shall be a minimum amount of staining on the paper.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurements).

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth put up as specified shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP–P–51.

6. NOTES

6.1 Intended use. The cotton flannels covered by this specification are intended for use in the manufacture of pajamas, gloves, the back of gloves, glove lining, padding for front interlining in coats and interlining for caps. Some of the specific uses of the cloths covered in this specification are as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

Type I, class 1 (shrunk)—Interlining for caps in MIL–C–15507.

Type 1, class 1—Interlining for caps in MIL–C–4233.

Type II, class 1—Padding for interlining of coats in MIL–C–27438.

—Padding for coat in MIL–C–19516.

—Chest padding for coats in MIL–C–15065.

Type III (dyed), class 3 (shrunk)—Pajamas in MIL–P–10961.

Type IV, class 2—Glove lining in JJ–G–451.

Type IV, class 3—For back, palm, and thumb for women's cotton work gloves in MIL–G–1008.

Type IV, class 3 (singed on unnapped side)—Gloves in MIL–G–4244.

Type IV, class 5—For back of gloves in JJ–G–451.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and class of cloth required (see 1.2).

(*c*) Color of cloth required (see 3.3).

(*d*) Width of cloth required (see 3.4.2).

(*e*) When singeing is required for type IV, class 3 (see 3.5.2.).

(*f*) When shrunk cloth for type I, class 1, type III (bleached) class 3, type III (dyed) class, is required (see 3.6).

(*g*) When length of piece is other than specified (see 3.7).

(*h*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP–P–51, may be exercised in the preparation for the delivery of cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Comparison of types and classes as pertains to CCC–C–458, CCC–F–466, and MIL–C–2324 is as follows:

CCC–C–458	*CCC–C–458a*
Type I	Type IV, class 1
Type II	Type III (bleached), class 2
Type III	Type II, class 1
Type IV	Type IV, class 5

CCC–F–466

Type I	Type III (bleached or dyed), class 1
Type II	Type III (bleached or dyed), class 2
Type III	Type III (bleached or dyed), class 3

MIL–C–2324	*CCC–C–458a*
Type I	Deleted
Type II	Deleted
Type III	Type III (bleached or dyed), class 3
Type IV	Deleted
Type V	Deleted
Type VI	Deleted

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

6.5 Supersession data. This specification includes, in part, the requirements of MIL–C–2324, dated July 24, 1950.

6.6 Transportation description. The transportation description and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indexed by name.

Carload minimum weight, any quantity.

Motor:

Cotton piece goods, not otherwise indexed, in original piece.

Truckload minimum weight, any quantity.

MILITARY CUSTODIANS:

Army—GL

Navy—SA

Air Force—11

Review activities:

Army—MD

Navy—SA

Air Force—11, 67

User activities:

Army—MO

Navy—None

Air Force—None

DSA—DM

Preparing activity:

Army—GL

Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing Of DOD Standardization Documents.

Copies of this specification may be purchased for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.